

Engineering Design & Architecture Specifications

System Overview: The platform utilizes a microservices architecture built on FastAPI backend systems and deployed via Docker containers.

Data Storage: Core transactional data is stored in PostgreSQL. Embeddings are generated using Sentence Transformers and cached in FAISS.

Security: Microservices communicate using gRPC with TLS. Authentication is managed via OAuth2 with JSON Web Tokens (JWT).

Scalability: Horizontal scaling is managed using Kubernetes. Auto-scaling rules are based on CPU usage exceeding 75% for more than 5 minutes.